

CSE 207: Data Structure and Algorithms - II



Dr. Mohammed Eunus Ali

Professor

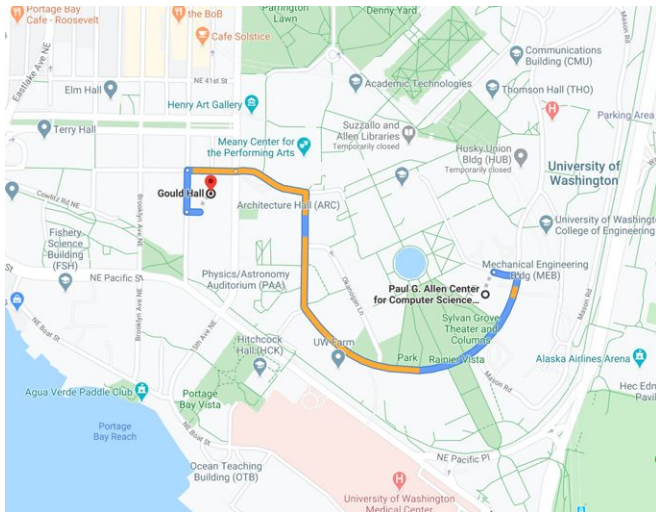
CSE, BUET

Course Teachers & Textbook

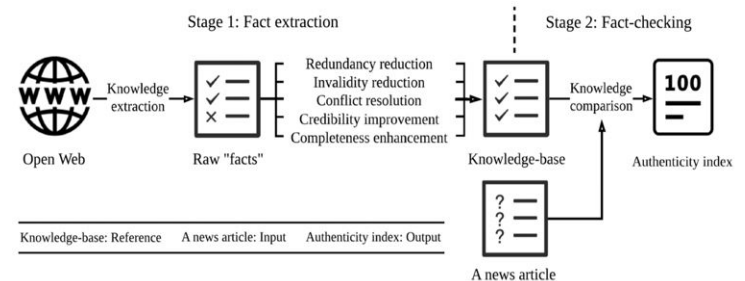
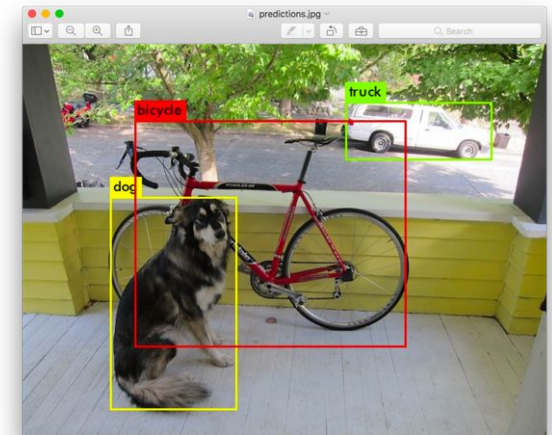
- Instructors
 - Dr Mohammed Eunus Ali (Part A)
 - Dr Sadia Sharmin (Part B)
- Resources
 - Slides
 - INTRODUCTION TO ALGORITHMS (3rd Edition)
 - Cormen, Leiserson, Rivest, Stein
 - Data Structures and Algorithms
 - Goodrich, Tamassia

Why 207?

1. Build a strong foundation of data structures and algorithms that will let you tackle the biggest problems in computing



207 Data Structures & Algorithms II



Why 208?

2. Pick up the advanced vocabulary, skills, and practice needed to make **design decisions**. Learn to **evaluate** the tools in your CS toolbox



What will we study?

- Advanced Graph Algorithms
 - Minimum Spanning Trees (MST), Shortest Paths etc.
 - Maximum flow and maximum bipartite matching, etc.
- Advanced Data Structures
 - Balanced binary search trees -AVL trees
 - Red-black trees, splay trees etc
 - Advanced heaps (Fibonacci heaps, binomial heaps)
- Algorithmic Theory
 - Lower bound theory
 - NP-Hard, NP-Completeness
- Complex algorithm design
 - Coping with hardness: Backtracking, branch and bound
 - Approximate Algorithms

Data Structures & Algorithms

- Data Structure:
 - A way of organizing, storing, accessing, and updating data
 - **Examples:** Arrays, Linked Lists, Stacks, Queues, Trees, Heaps
- Algorithm:
 - A series of precise instructions to produce a specific outcome
 - **Examples:** Binary Search, Backtracking, Greedy, Dynamic
- Program:
 - A program is the expression of an algorithm in a programming language

Data Structure + Algorithms
Graph + Breadth First Search

Correctness

- How do you know an algorithm is correct?
 - For every input instance, it halts with the correct output
 - Since there are usually infinitely many inputs, it is not trivial

Efficiency

- Correctness alone is not sufficient
- Brute-force algorithms exist for most problems
- To sort n numbers, we can enumerate all permutations of these numbers and test which permutation has the correct order
 - Why cannot we do this?
 - Too slow!
 - By what standard?

Why Study Algorithms and Data Structure

- You will write better, faster, more elegant code.
- You will think more clearly, more abstractly and more mathematically.
- You will be able to solve new problems.
- You will be able to give non-trivial methods to solve problems.
- You will improve your research skills in almost any area.
- It's one of the most challenging and interesting area of Computer Science.

Why Study Algorithms and Data Structure

- Almost all big companies want programmers with knowledge of algorithms: Microsoft, Apple, Google, Facebook, Oracle, IBM, Yahoo, NIST etc.
- In most programming job interviews, they will ask you several questions about algorithms and/or data structures. They may even ask you to write pseudo or real code on the spot.
- Your knowledge of algorithms will set you apart from the masses of interviewees who know only how to program.

.

Course Outline (Part I)

- Week 1
 - Introduction to Graph Algorithms
- Week 2
 - MST, Shortest Paths
- Week 3
 - All Pair Shortest Paths
- Week 4
 - Maximum flow and maximum bipartite matching
- Week 5
 - Balanced binary search trees -AVL trees
- Week 6
 - Red-black trees, splay trees etc.
- Week 7
 - Advance Heaps



The End